

Interior Point Methods

Recall application to LP

$$\min: c^T x$$

$$\text{st: } Ax \leq b$$

$\uparrow$
 $m \times n$
 $x \in \mathbb{R}^n$



$$F_t(x) = t \cdot c^T x + F_0(x)$$

$$= t \cdot c^T x + - \sum_{i=1}^m \log(b_i - a_i^T x)$$

Barrier function



$$x_t \in \arg \min: F_t(x)$$

Let $t \rightarrow \infty$: 1st term dominates

$x_t \rightarrow x^* = \text{solution original LP}$

- $\rightarrow$
- ① Need to minimize $F_t(x)$ quickly
(Newton) $\rightarrow$ very fast when we start in region of quad. conv.
 - ② Increase t : $t^+ = (1+\alpha)t$ — geometric increase
But must have x_t in region of quad conv of F_{t^+}
 - ③ $c^T x_t - c^T x^* \leq O(1/t)$

Problem: $-\log x$

Need to minimize: $F_t(x) = t \cdot c^T x - \sum \log(b_i - a_i^T x)$

But we learned: Newton's method needs

① F_t strongly convex; smooth

② Lipschitz Hessian: $\|\nabla^2 F_t(x) - \nabla^2 F_t(y)\|_2 \leq L \cdot \|x - y\|_2$

Does $-\log x$ satisfy this?

$$f(x) = -\log x,$$

$$\nabla f(x) = -\frac{1}{x}$$

$$\nabla^2 f(x) = \frac{1}{x^2}$$

$$\rightarrow \left\| \frac{1}{x^2} - \frac{1}{y^2} \right\| \leq L \cdot |x - y|$$

LP takes opt values on bdy, so we cannot keep x away from 0.

Affine Invariance of Newton's Method

Unlike Gradient Descent, Newton's method is affine invariant.

$\min: f(x)$ $y = Ax \rightarrow \boxed{g(y) = f(Ay)}$ then compare the
trajectory of Newton's method on $f(x): x_0, x_1, x_2, \dots$
 $g(y): y_0, y_1, y_2, \dots$

Conditions required for Newton If: $y_0 = Ax_0 \Rightarrow y_k = Ax_k$

Hessian Lipschitzness: $\|\nabla^2 f(x) - \nabla^2 f(y)\|_2 \leq L \|x - y\|_2$

Check, for example, that this condition changes for
 $f(x) = \log x$.

Affine Invariance of Conditions?

$\nabla^2 f$ Lipschitz: $\|\nabla^2 f(x) - \nabla^2 f(y)\|_2 \leq L \|x - y\|_2$

Take directional derivative:

$$\|\nabla^2 f(x + \varepsilon \cdot u) - \nabla^2 f(x)\|_2 \leq L \cdot \varepsilon \cdot \|u\|_2$$

as $\varepsilon \rightarrow 0$: $\boxed{\|\nabla^3 f(x)[u]\| \leq L \|u\|_2}$

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (f(x + \varepsilon u) - f(x)) = \nabla f(x)^T u = \nabla f(x)[u]$$

$\uparrow$
vector: linear map from $\mathbb{R}^n \rightarrow \mathbb{R}$

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (\nabla f(x + \varepsilon u) - \nabla f(x)) = \nabla^2 f(x) u = \nabla^2 f(x)[u]$$

$\uparrow$
matrix: bilinear map from $\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (\nabla^2 f(x + \varepsilon u) - \nabla^2 f(x)) = \nabla^3 f(x)[u]$$

$$\nabla^2 f \text{ is Lipschitz: } \|\nabla^3 f(x)[u]\|_2 \leq L \cdot \|u\|_2 \quad \forall x$$
 $\Leftrightarrow$

(check - recalling that $[\nabla^3 f(z)[u]]$ is $n \times n$ matrix)

$$\|\nabla^3 f(x)[u, v, v]\| \leq L \cdot \|u\|_2 \cdot \|v\|_2^2$$

↑
affine invariant

not affine invariant

We will replace $\|\cdot\|_2$ with an affine-invariant norm.

An Affine Invariant Local Norm

Recall that we can define: $\|v\|_Q^2 = v^T Q v$

(if $Q = I$ then this is the Euclidean norm)

We use Q that depends on x : $Q = \nabla^2 f(x)$

$$\|u\|_{\nabla^2 f(x)}^2 = u^T [\nabla^2 f(x)] u = \nabla^2 f(x)[u, u]$$

Notation: $\|u\|_x^2 \triangleq \|u\|_{\nabla^2 f(x)}^2$

We will use: $f(x) = F_t(x) = t C^T x + F_0(x)$

$$\nabla^2 F_t = \nabla^2 F_{t'} \quad \forall t, t'$$

Self Concordant Functions

Def'n: A function f is called M -self-concordant

if: $\underbrace{\|\nabla^3 f(x)[h, h, h]\|}_{\text{compare to what had previously:}} \leq 2M \cdot \|h\|_x^3$

compare to what had previously: $\|\nabla^3 f(x)[h, h, h]\| \leq L \cdot \|h\|_2^3$.

Example: $f(x) = -\log(x)$

$$\begin{array}{l|l} \nabla f(x) = -\frac{1}{x} & \nabla^3 f(x)[h, h, h] = -\frac{2x^{-3}h^3}{1} \\ \nabla^2 f(x) = \frac{1}{x^2} & \|h\|_x^2 = h^T \nabla^2 f(x) h = \frac{h^2}{x^2} \\ \nabla^3 f(x) = -\frac{2}{x^3} & \end{array}$$

We have: $-\log x$ is 1-self-concordant

$$|\nabla^3 f(x)[h, h, h]| \leq 2 \cdot 1 \cdot \|h\|_x^3$$

Self concordance is affine invariant

Function $f(x)$. Assume f is M -self-concordant.

$$y = Ax, \quad x = A^{-1}y$$

$$\text{Define: } g(y) = f(Ax), \quad v = Au$$

$$\nabla g(y)[u] = \nabla f(x)[v]$$

$$\nabla^2 g(y)[u, u] = \nabla^2 f(x)[v, v]$$

$$\nabla^3 g(y)[u, u, u] = \nabla^3 f(x)[v, v, v]$$

$\Rightarrow g$ is M -self-concordant.

(check the details)